

PAPER_242 — Rings of Relativity: Einstein Ring Lensing Amplification in the Full MUGE

Author: Daniel T. Murphy

Version: 1.0 | Session: 60 | PAPER Number: 242

PAPER_242 — Rings of Relativity: Einstein Ring Lensing Amplification in the Full MUGE

GAL-CLUS-022058s — Static Einstein-Ring Lensing Factor in the Master Universal Gravity Equation

Author: Daniel T. Murphy **Framework:** Unified Quantum Field Framework (UQFF) v4.10 **Calculator Class:** RingsOfRelativityEinsteinLensingMUGECalculator (CP3, Session 60) **Encoded By:** Grok (xAI), October 2025 (C++ source); Python CP3 integration March 2026 **Version:** 1.0 | **Session:** 60 | **PAPER Number:** 242

<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $M_{\text{UQFF}} = 1.43e1 \text{ TeV}$ -->

Abstract

This paper presents the full Master Universal Gravity Equation (MUGE) for the GAL-CLUS-022058s gravitational lens system ("Rings of Relativity"). The key unique mathematical contribution is the **static Einstein ring lensing amplification factor** L_t , derived from the angular diameter distance ratio D_{LS}/D_S and the Schwarzschild radius of the galaxy cluster lens, applied as a multiplicative correction to the base gravitational field.

This is expressly **distinct** from the dynamic lensing modulation $L(t) = L_0 e^{-t/\tau} \cos(\omega t)$ already captured in CP3 class 81 (UQFFLensingModulationRingsCalculator, Session 53). The present paper captures the static, geometry-driven Einstein ring correction.

2. System Parameters

Parameter	Symbol	Value	Units
Lens mass	M	$10^{14} M_{\odot}$	kg
Einstein radius	r	3.086×10^{20}	m (~10 kpc)
Redshift of lens	z_{lens}	0.5	dimensionless
Angular distance ratio	$L_{\text{factor}} = D_{\text{LS}}/D_S$	0.67	dimensionless
Magnetic field	B	10^{-5}	T
Gas velocity	v_{gas}	10^5	m/s
Oscillatory amplitude	A_{osc}	10^{-12}	m/s^2

3. The Novel Lensing Amplification Term

3.1 Einstein Ring Lensing Factor

The static gravitational lensing amplification of the measured gravitational field along the line of sight through the Einstein ring geometry:

$$L_t = \frac{GM}{c^2 r} \cdot L_{\text{factor}}, \quad \text{quad } L_{\text{factor}} = \frac{D_{\text{LS}}}{D_S} = 0.67$$

$$\text{corr}_L = 1 + L_t$$

where:

$GM/c^2 r$ is the dimensionless Schwarzschild-radius-to-Einstein-radius ratio

D_{LS} = angular diameter distance from lens to source

D_S = angular diameter distance from observer to source

The factor D_{LS}/D_S encodes the lensing geometry in the single-lens single-source approximation

3.2 Friedmann Correction at $z = 0.5$

$$H(z=0.5) = H_0 \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda} = H_0 \sqrt{0.3 \cdot (1.5)^3 + 0.7}$$

This mid-redshift Friedmann correction is distinct from $z=0$ (local), $z=3.5$ (HUDF), or $z=0.01$ (nearby lenses) used in earlier classes.

4. Full MUGE: All Nine Terms

$$g_{\text{Rings}}(r, t) = \underbrace{T_1}_{\text{base+H+B+L}} + \underbrace{T_2}_{\text{UQFF}} + \underbrace{T_3}_{\Lambda} + \underbrace{T_4}_{\text{EM}} + \underbrace{T_q}_{\text{QM}} + \underbrace{T_{\text{fl}}}_{\text{fluid}} + \underbrace{T_{\text{osc}}}_{2\text{-mode}} + \underbrace{T_{\text{DM}}}_{\text{DM+pert2}} + \underbrace{T_{\text{wind}}}_{\text{stellar, wind}}$$

Term 1 — Base gravity with $H(z)$, magnetic, Einstein lensing:

$$T_1 = \frac{GM}{r^2} (1 + H(z), t) (1 - B/B_{\text{crit}}) (1 + L_t)$$

Term 2 — UQFF unified field with time-reversal:

$$T_2 = (U_{g1} + U_{g4}) (1 + f_{\text{TRZ}})$$

Term 3 — Cosmological constant:

$$T_3 = \frac{\Lambda c^2}{3}$$

Term 4 — Electromagnetic with vacuum density ratio:

$$T_4 = \frac{q v_{\text{gas}} B}{m_p} \left(1 + \frac{\rho_{\text{UA}}}{\rho_{\text{SCM}}} \right) \cdot s_{\text{EM}}, \quad \text{quad } \frac{\rho_{\text{UA}}}{\rho_{\text{SCM}}} = \frac{7.09 \times 10^{-36}}{7.09 \times 10^{-37}} = 10$$

Term 5 — Quantum uncertainty:

$$T_q = \frac{\hbar}{\sqrt{\Delta x, \Delta p}}, \quad \psi, \quad \frac{2\pi}{t_H}$$

Term 6 — Fluid (buoyancy-acceleration):

$$T_{\text{fl}} = \frac{\rho_{\text{fluid}}}{\rho_{\text{base}}} \cdot V, \quad g_{\text{base}} \{M\}$$

Term 7 — Two-mode oscillation (standing wave + traveling wave):

$$T_{\text{osc}} = 2A \cos(kx) \cos(\omega t) + \frac{2\pi}{t_{\text{Gyr}}} A \cos(kx - \omega t)$$

The first mode is a **standing wave decomposition** ($2\cos\cos$); the second mode is a **Gyr-scaled traveling wave**. Together they form a beating-mode pair unique to this system.

Term 8 — Dark matter with $\delta_2 = 3GM/r^3$ perturbation:

$$T_{\rm DM} = \frac{(M + M_{\rm DM})(\delta\rho/\rho + 3GM/r^3)}{M}$$

The second-order perturbation $3GM/r^3$ is a tidal-force density correction distinct from density-contrast $\delta\rho/\rho$.

Term 9 — Stellar wind feedback:

$$T_{\rm wind} = \frac{\rho_{\rm wind} v_{\rm wind}^2}{\rho_{\rm fluid}}$$

5. Distinction from Prior Art

Feature	Class 81 (Session 53)	This class (Session 60)
Lensing factor	$L(t) = L_0 e^{-t/\tau} \cos(\omega_{\rm lens} t)$	$L_t = (GM/c^2 r) \cdot (D_{\rm LS}/D_{\rm S})$
Physics	Dynamic lens transit alignment	Static Einstein ring geometry
Time dependence	Yes (decaying oscillation)	No (geometric constant)
Parameter	$L_0=0.15, \tau_{\rm lens}, \omega_{\rm lens}$	$L_{\rm factor}=0.67$ (distance ratio)
Oscillation	None	Two-mode standing + traveling wave
DM pert2	Not present	$3GM/r^3$ tidal correction
Wind term	Not present	$\rho_w v_w^2 / \rho_{\rm fl}$

6. Numerical Verification

Using default parameters ($M = 1.989 \times 10^{44}$ kg, $r = 3.086 \times 10^{20}$ m, $z=0.5$):

$$L_t = \frac{(6.6743 \times 10^{-11})(1.989 \times 10^{44})}{(2.998 \times 10^8)^2 (3.086 \times 10^{20})} \times 0.67 \approx 1.6 \times 10^{-3}$$

$${\rm corr}_L \approx 1.0016 \quad (\sim 0.16\%, {\rm amplification})$$

$$H(z=0.5) = H_0 \sqrt{0.3 \times 3.375 + 0.7} \approx 1.27 H_0$$

The lensing correction is small but non-zero and physically motivated by the Einstein ring geometry of GAL-CLUS-022058s ($\theta_E \approx 10''$, confirmed by HST imaging).

7. Integration Into UQFF Pipeline

CP3 class: RingsOfRelativityEinsteinLensingMUGECalculator (112th calculator, Session 60)

Aggregator: v2.4.0 — registered in CP3_CALCULATORS

Source: Doc 8 C++ class RingsOfRelativity (Grok/xAI, October 2025)

Complementary classes:

Class 81 UQFFLensingModulationRingsCalculator — dynamic lens transit

Class 111 NGC3603FullMUGECavityPressureCalculator — companion new class (PAPER_243)

8. References

Hubble Space Telescope imaging of GAL-CLUS-022058s (Infante et al. 2022)

Einstein (1936): Lens-star problem, Science 84, 506

Schneider, Ehlers & Falco (1992): Gravitational Lenses, Springer

Murphy, D.T. (2025): UQFF Manuscript — Doc 8 (Rings of Relativity MUGE, October 2025)

Grok (xAI): C++ class RingsOfRelativity, encoded October 2025

PAPER_242 | Session 60 | CP3 class 111 (RingsOfRelativityEinsteinLensingMUGECalculator) | UQFF v4.10

PAPER_242 | UQFF v4.74 | Star-Magic UQFF © 2025 Daniel T. Murphy